

PATTANI, Thailand

WMO: 485800

Lat: 6.7897N

Lon: 101.1469E

Elev: 20

StdP: 14.69

Time Zone: 7.00 (E07)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	70.7	72.1	66.7	98.3	80.0	67.9	102.7	79.8	12.1	84.3	11.2	84.5	0.1	120	0.299

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	15.1	94.9	79.5	93.7	79.3	92.7	79.2	81.8	91.1	81.0	90.2	80.6	89.7	5.8	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.2	151.9	86.6	78.7	149.3	85.9	78.2	146.5	85.3	45.6	91.6	44.8	90.6	44.3	90.1	89.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years					
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
11.6	9.9	8.5	DB	68.2	96.8	2.6	1.9	66.3	98.1	64.8	99.2	63.3	100.3	61.4	101.7
			WB	66.4	83.4	3.3	1.8	64.0	84.8	62.1	85.9	60.2	86.9	57.8	88.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	82.5	80.1	81.1	82.8	84.4	84.7	84.0	83.4	83.3	82.9	82.0	80.9	80.0
	DBStd	2.29	1.78	1.72	1.73	1.64	1.66	1.44	1.51	1.55	1.35	1.59	2.06	2.08
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11853	934	872	1016	1032	1074	1021	1035	1032	986	994	927	930
	CDD65	6378	469	452	551	582	609	571	570	567	536	529	477	465
	CDH74	64591	4071	4329	5818	6700	6890	6290	6084	5950	5360	4980	4258	3861
(13)	CDH80	25406	1339	1745	2573	3094	3037	2672	2507	2442	2014	1670	1251	1062
(14)	Wind	WSAvg	1.8	2.1	2.2	2.1	1.8	1.6	1.8	2.0	1.8	1.6	1.0	1.6
(15) Precipitation	PrecAvg	75.6	3.0	1.5	2.5	3.1	5.1	4.4	4.7	5.7	5.4	9.1	15.5	15.6
	PrecMax	102.0	13.0	11.0	10.8	9.5	11.7	10.1	8.3	12.7	8.4	18.7	34.9	38.1
	PrecMin	53.5	0.1	0.0	0.0	0.0	1.1	0.6	1.7	2.9	2.0	3.9	6.0	1.5
	PrecStd	14.6	3.3	2.5	2.7	2.3	2.2	2.0	1.8	2.4	1.5	3.0	7.2	9.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.2	93.0	95.2	96.9	96.4	94.7	94.1	94.1	92.9	92.1	90.8	90.4
		MCWB	77.9	77.8	78.7	80.4	79.7	79.6	78.6	78.5	79.1	79.4	79.2	78.7
	2%	DB	89.3	91.3	93.4	95.0	94.5	93.3	92.5	92.7	91.5	90.6	89.4	88.8
		MCWB	77.5	77.5	78.8	79.9	80.0	79.5	78.8	78.7	79.1	79.1	79.1	78.2
	5%	DB	87.9	90.0	92.1	93.5	93.0	92.0	91.4	91.4	90.4	89.3	88.1	87.5
		MCWB	77.0	77.3	78.5	79.6	79.9	79.4	78.7	78.7	78.9	78.8	78.8	77.9
	10%	DB	86.5	88.6	90.7	92.0	91.6	90.7	90.1	90.0	89.0	87.9	86.6	85.8
		MCWB	76.6	76.9	78.2	79.4	79.7	79.2	78.6	78.5	78.6	78.6	78.6	77.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.0	80.5	81.9	82.8	82.5	82.3	81.5	81.5	81.9	81.0	80.9	80.4
		MCDB	87.9	89.9	91.9	93.4	92.4	91.1	90.2	90.2	89.6	88.7	88.0	87.3
	2%	WB	78.9	79.3	80.5	81.4	81.5	81.0	80.4	80.4	80.5	80.1	80.1	79.4
		MCDB	86.8	88.6	90.6	91.8	91.2	89.8	89.3	89.4	88.6	87.7	86.9	86.1
	5%	WB	78.2	78.4	79.7	80.6	80.8	80.4	79.8	79.7	79.7	79.5	79.5	78.7
		MCDB	85.5	87.4	89.4	91.0	90.3	89.1	88.5	88.5	87.7	86.9	86.0	85.1
	10%	WB	77.4	77.6	78.9	79.9	80.1	79.7	79.1	79.0	79.0	78.9	78.9	77.9
		MCDB	84.2	86.2	88.4	89.7	89.1	88.3	87.7	87.7	87.0	86.2	85.1	83.7
(35) Mean Daily Temperature Range	5% DB	MDBR	13.2	16.0	16.8	16.3	15.1	14.7	14.6	14.7	13.7	12.5	10.7	10.6
		MCDBR	15.4	17.7	18.8	18.3	16.7	16.1	16.1	16.2	15.2	14.6	13.5	13.7
		MCWBR	5.5	6.1	6.3	5.9	5.5	5.4	5.4	5.5	5.3	5.4	5.0	5.2
	5% WB	MCDBR	13.4	15.8	16.9	16.7	15.7	15.2	15.3	15.6	14.5	13.4	12.2	12.0
		MCWBR	5.5	6.0	6.4	6.0	5.7	5.7	5.8	6.0	5.7	5.6	5.2	5.3
(40) Clear-Sky Solar Irradiance	taub		0.461	0.461	0.470	0.457	0.450	0.461	0.465	0.463	0.454	0.453	0.453	0.465
	taud		2.297	2.272	2.270	2.310	2.353	2.350	2.329	2.321	2.340	2.344	2.349	2.299
	Ebn at Noon		272	275	272	272	268	261	261	266	273	274	272	268
	Edh at Noon		43	44	45	43	40	39	40	42	41	41	40	42
(44) All-Sky Solar Radiation	RadAvg		1492	1807	1860	1868	1668	1596	1579	1639	1602	1468	1293	1248
(45)	RadStd		112	117	153	83	91	64	78	52	57	100	110	138

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	+1.15	+0.49	+0.56	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)	+0.36	N/A	N/A	+0.54	+0.45	+0.34	N/A	N/A	+140	+142